Sheet 03: Isometric Scales and Isometric Views

TOPIC: ISOMETRIC PROJECTION VS ISOMETRIC VIEW

1. MATHEMATICAL BASIS:

- Isometric Length = True Length x cos(45) / cos(30) = sqrt(2/3) x True Length "H
- Isometric View is drawn using true dimensions along 30-degree isometric axes.
- Isometric Projection is drawn using the reduced isometric scale (81.6%).

2. FOUR-CENTER METHOD FOR ISOMETRIC CIRCLES (ELLIPSES):

Step 1: Enclose the circle of diameter D in an isometric rhombus of side D.

Step 2: From obtuse corners, draw lines to midpoints of opposite sides.

Step 3: Use intersection points as centers to draw 4 connecting circular arcs.

3. ASSEMBLY TOLERANCES & FITS (ISO 286):

- Clearance Fit: Hole minimum size > Shaft maximum size.
- Transition Fit: Tolerance zones overlap.
- Interference Fit: Shaft minimum size > Hole maximum size.

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